NUST

NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

Target Based Design

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Learning Objectives

1. To Understand the methods in protein modeling

2. To Understand evaluation strategies for protein modeling

3. To Learn development and evaluation of homology model

Protein Structure

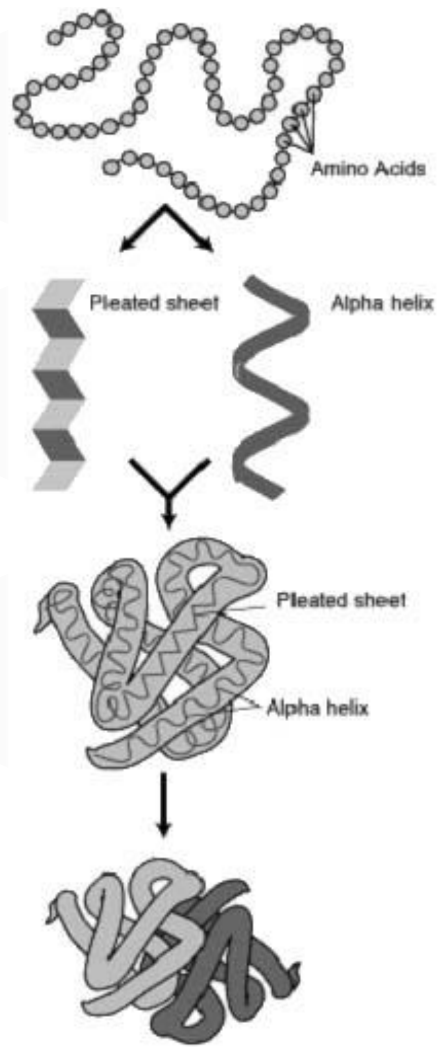
STRUCTURE

Primary — Assembly

Secondary — Folding

Tertiary — Packing

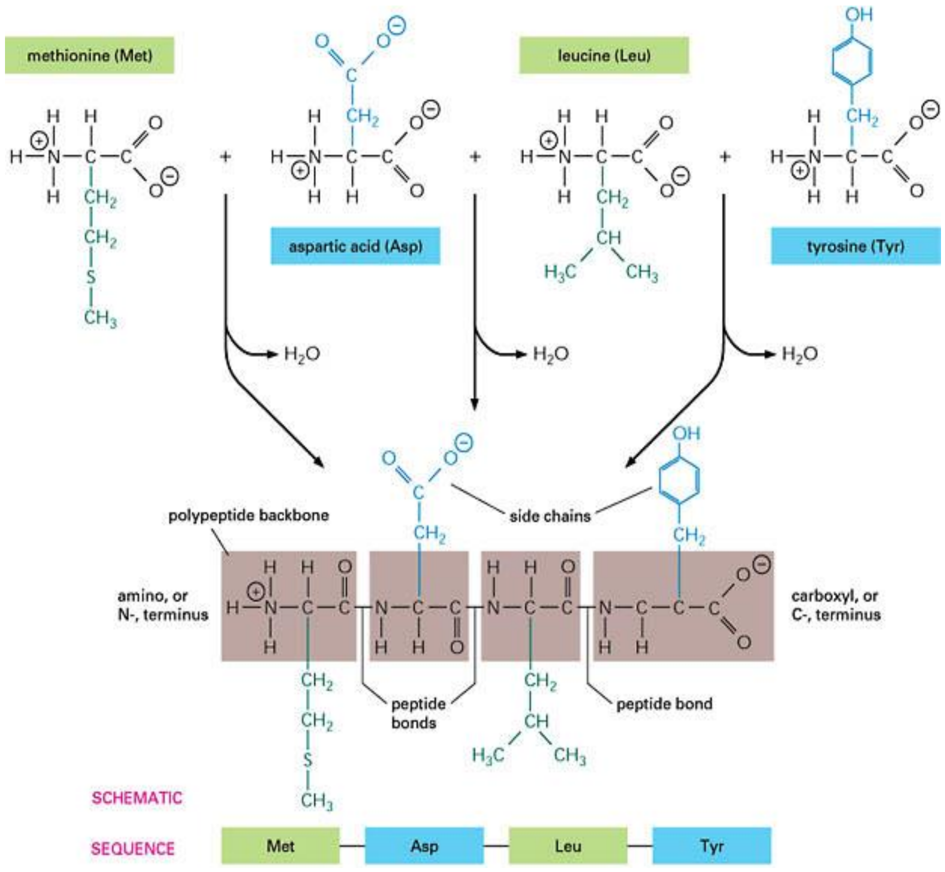
Quaternary — Interaction



PROCESS

Protein Assembly

- Occurs at the ribosome
- Involves dehydration synthesis and polymerization of amino acids attached to tRNA:
- Thermodynamically unfavorable
- Yields primary structure



Primary Structure

- Linear
- Ordered
- 1 dimensional
- Sequence of amino acid polymer
- By convention, written from amino end to carboxyl end
- A perfectly linear amino acid polymer is neither functional nor energetically favorable → folding!

Protein Folding

- Involves localized spatial interaction among primary structure elements, i.e. the amino acids
- It reduce ΔE (thermo-dynamically favorable)
- Yields secondary structure

Example

- **Ribonuclease**, a small protein that contains **8 cysteins linked via four disulfide bonds**
- **Urea** in the presence of **2-mercaptoethanol** fully **denatures ribonuclease**
- When **urea and 2-mercaptoethanol** are **removed**, the protein spontaneously **refolds**, and the correct disulfide **bonds are reformed**
- Quite **"simple"** experiment, but so important it earned Chris Anfinsen the 1972 Chemistry Nobel Prize

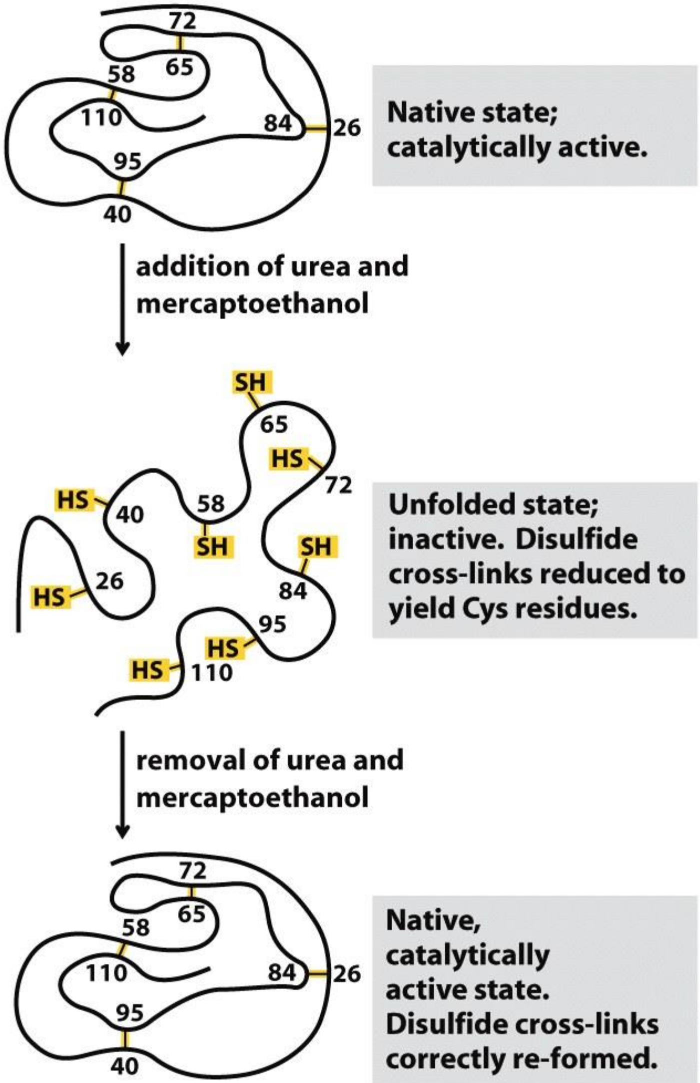


Figure 4-26
Lehninger Principles of Biochemistry, Fifth Edition

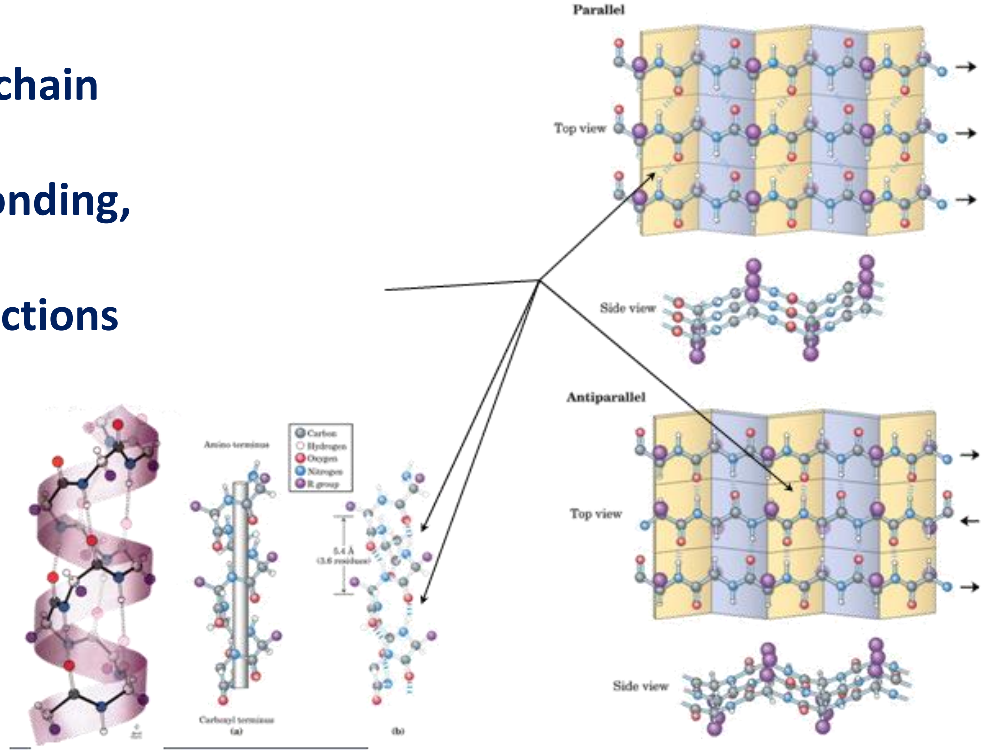
Secondary Structure

Non-linear
3 dimensional

Localized to regions of an amino acid chain

Formed and stabilized by hydrogen bonding,

Electrostatic and van der Waals interactions



The α helix

Helical backbone is held together by hydrogen bonds between the nearby backbone amides

Peptide bond has a strong dipole moment
- Carbonyl O negative
- Amide H positive

Negatively charged residues often occur near the positive end of the helix dipole

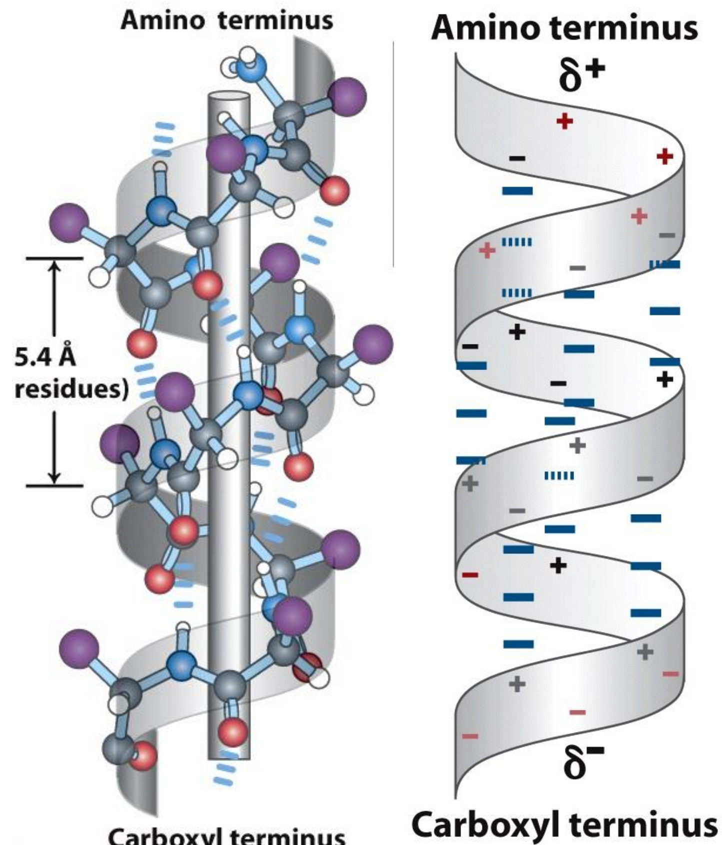


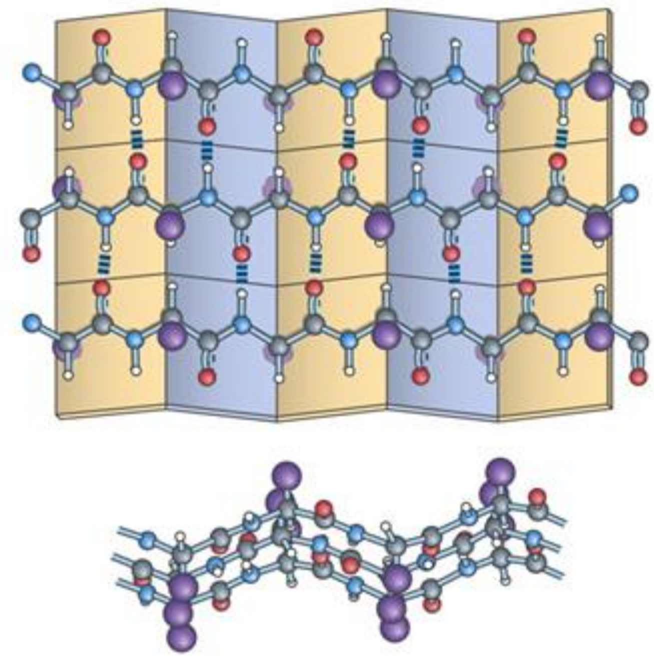
Figure 4-5
Lehninger Principles of Biochemistry, Fifth Edition

β Sheets

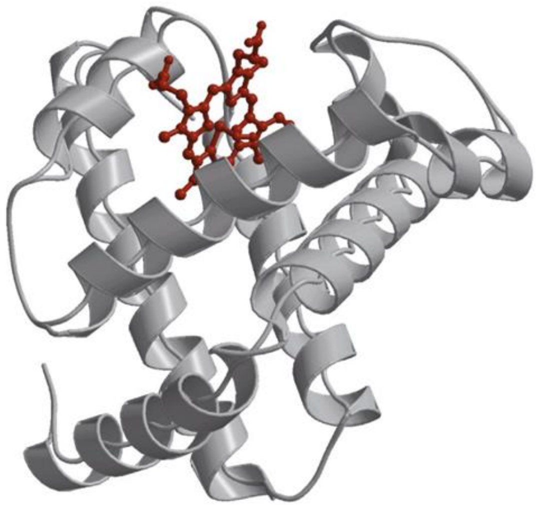
The planarity of the peptide bond and tetrahedral geometry of the α-carbon create a pleated sheet-like structure

Sheet-like arrangement of backbone is held together by hydrogen bonds between the more distal backbone amides

Side chains protrude from the sheet alternating in up and down direction



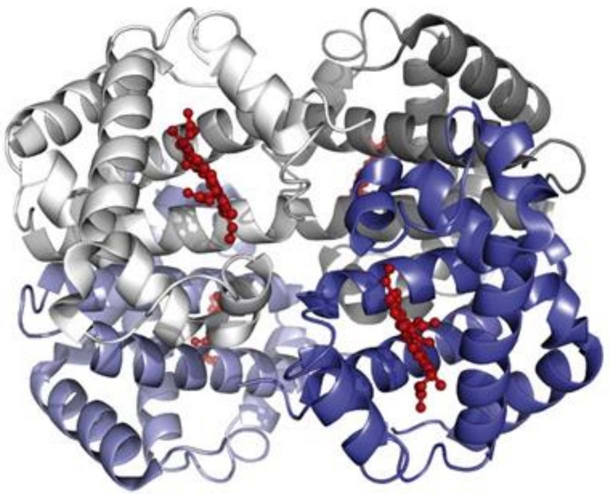
Protein Tertiary Structure



- The alpha-helices and beta-sheets are folded into compact globule units called protein domains

- Whole proteins can comprise one or several such domains

Quaternary Structure



Quaternary structure is formed by spontaneous assembly of individual polypeptides into a larger functional cluster

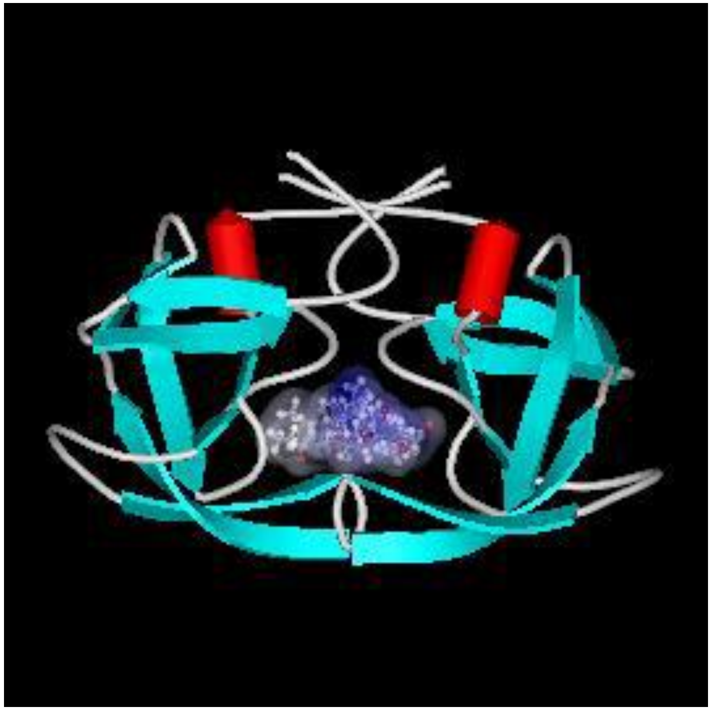
- hemoglobin
- ion channels

Structure based design

Uses information from the 3D structure of the protein

Protein 3D structure

- Homology modeling
- Fold recognition
- *ab initio calculations*



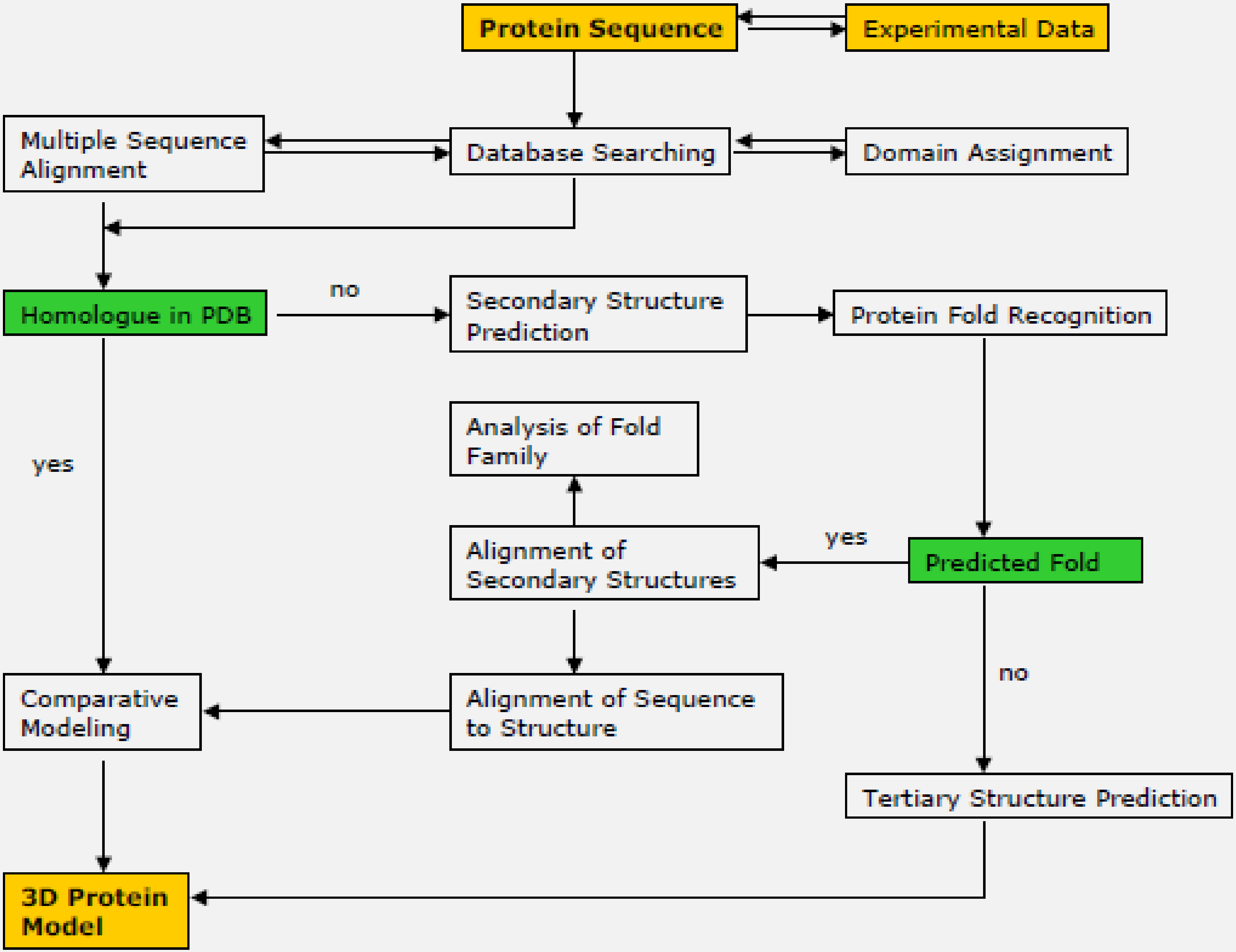
Protein Modeling

Homology Modeling = comparative modeling

For proteins with unknown 3D structure which have a similar sequence (>26%) as a protein with a known 3D structure .

Fold Recognition/Threading

For proteins which don‘t share sequence identity with a protein with known 3D structure.

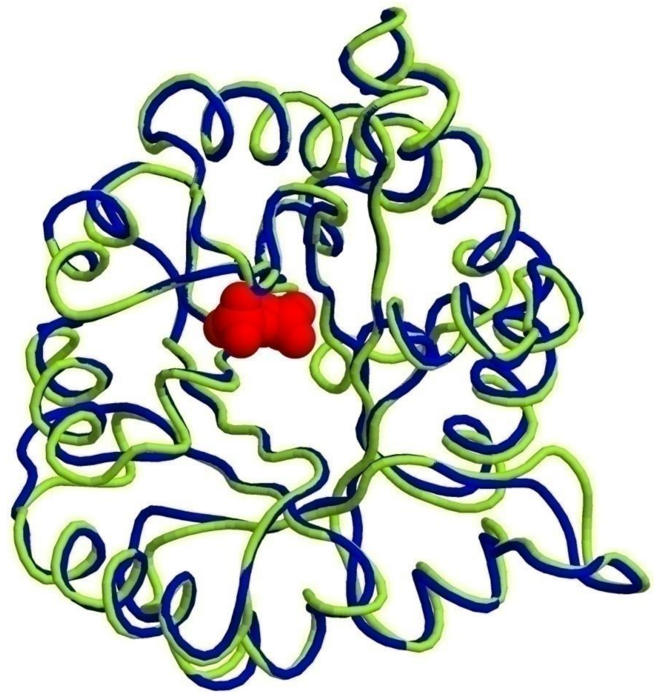

Protein Sequence
Experimental Data
Multiple Sequence Alignment
Database Searching
Domain Assignment
Homologue in PDB
no
Secondary Structure Prediction
Protein Fold Recognition
yes
Analysis of Fold Family
Alignment of Secondary Structures
yes
Predicted Fold
no
Comparative Modeling
Alignment of Sequence to Structure
Tertiary Structure Prediction
3D Protein Model

Homology Modeling Basic Idea

1. A protein structure is defined by its amino acid sequence.

2. Closely related sequences adopt highly similar structures, distantly related sequences may still fold into similar structures.

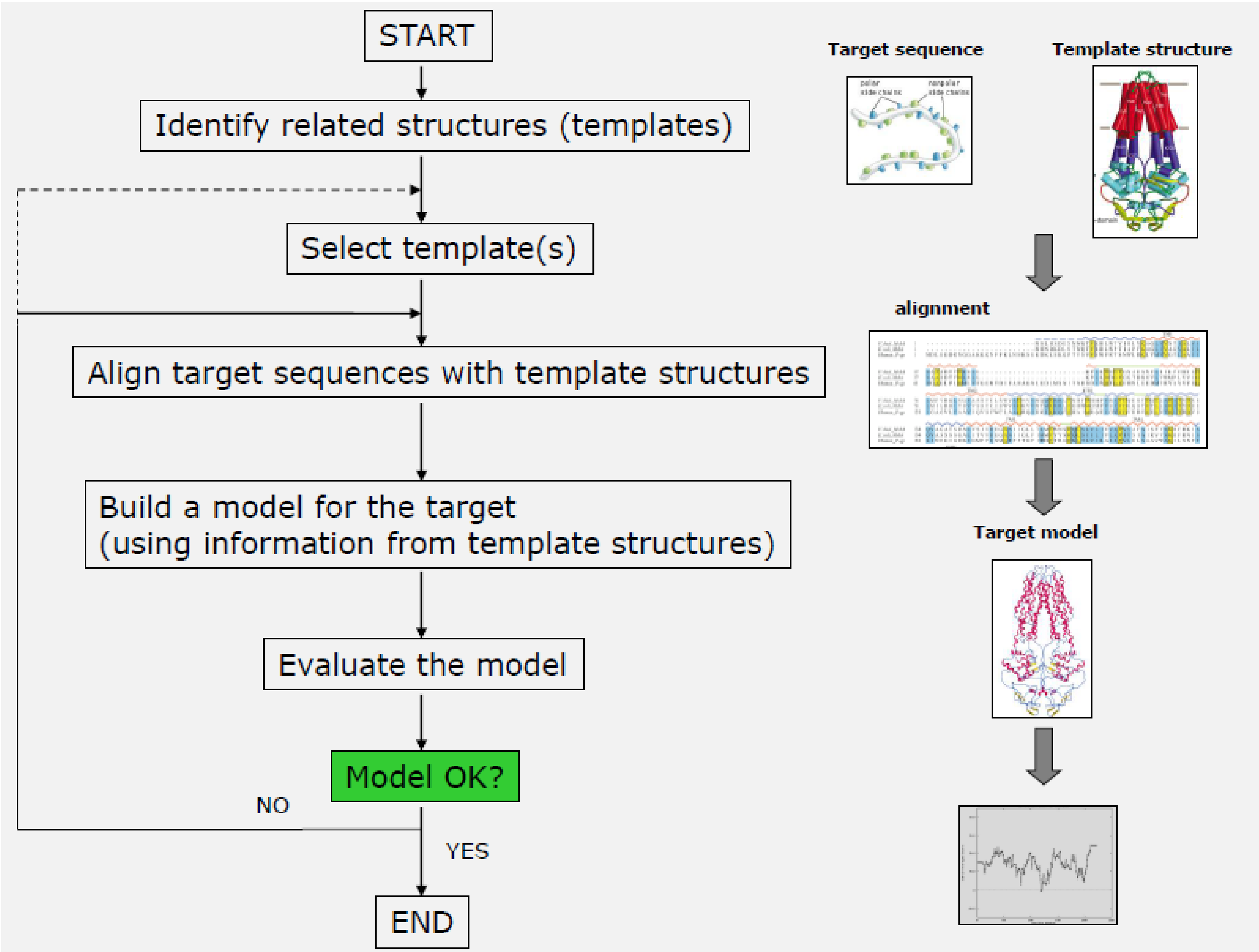
3. Three-dimensional structure of proteins from the same family is more conserved than their primary sequences.



Triophospate isomerases
44.7% sequence identity
0.95 RMSD

Steps in homology-modeling

1. Template search and initial sequence alignment
2. Alignment optimization
3. Generation of the target protein backbone according to the coordinates of the template
4. Modeling of missing variable regions (insertions, loops)
5. Modeling of the side chains
6. Model optimization (Energy minimization, molecular dynamics)
7. Model validation (Ramachandran plot, stereochemistry)


START
Identify related structures (templates)
Select template(s)
Align target sequences with template structures
Build a model for the target
(using information from template structures)
Evaluate the model
Model OK?
NO
YES
END
Target sequence
Template structure
alignment
Target model

FASTA File Format

http://www.uniprot.org/

Sequence representation:
FASTA format

UniProt entry code

gene name

short description
of function

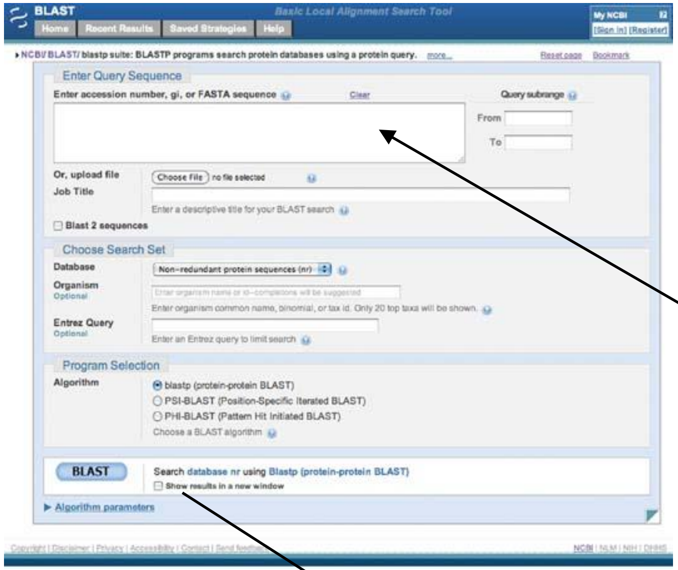
species

protein sequence
(single letter code)

```
>sp|P30531|SC6A1_HUMAN Sodium- and chloride-dependent GABA transporter 1
OS=Homo sapiens GN=SLC6A1 PE=1 SV=2
MATNGSKVADGQISTEVSEAPVANDKPKTLVVKVQKKAADLPDRDTWKGRFDF
LMSCVGY
AIGLGNVWRFPYLCGKNGGGAFLIPYFLTLIFAGVPLFLLECSLGQYTSIGGLGVW
KLAP
MFKGVGLAAAVLSFWLNIYYIVIISWAIYYLYNSFTTTLPWKQCDNPWNTDRCFS
NYSMV
NTTNMTSAVVEFWERNMHQMTDGLDKPGQIRWPLAITLAIAWILVYFCIWKGV
GWTGKVV
YFSATYPYIMLIILFFRGVTLPGAKEGILFYITPNFRKLSDSEVWLDAATQIFFSYGL
GL
GSLIALGSYNSFHNNVYRDSIIVCCINSCTSMFAGFVIFSIVGFMAHVTKRSIADVA
ASG
PGLAFLAYPEAVTQLPISPLWAILFFSMLLMLGIDSQFCTVEGFITALVDEYPRLLR
NRR
ELFIAAVCIISYLIGLSNITQGGIYVFKLFDYYSASGMSLLFLVFFECVSISWFYGVN
RF
YDNIQEMVGSRPCIWWKLCWSFFTPIIVAGVFIFSAVQMTPLTMGNYVFPKWGQ
GVGWLM
ALSSMVLIPGYMAYMFLTLKGSLKQRIQVMVQPSEDIVRPENGPEQPQAGSSTSK
EAYI
```

Template search

- BLAST search against PDB database (http://blast.ncbi.nlm.nih.gov/)
- Template should have at least 25% sequence identity
- Active or inactive state is important to consider?



paste sequence of protein with unknown tertiary structure (FASTA format)

BLAST your sequence against database of choice, i.e. PDB (to find out if close homologue has already been crystallized)

Alignment optimisation

For low sequence identity: multiple sequence alignment

```
A_1B0U (   30) visiigssgsgkstflrcinflekpsegaiivngqninlvrdkdgqlkvadknqlrllrtrl (91   )
A_1F2U (   25) inliiggngsgksslldailvglywplrikdikkdeftkvgardtyidlifekdgtkyritr (86   )
A_1F3O (   31) fvsi|gpsgsgkst|lniigcldkptegevyid---------niktndldddeltkirrdki (81   )
A_1G29 (   31) fmillgpsgcgktttlrmiagleepsrgqiyig---------dklvadpekgifvppkdrdi (83   )
   TAP (   45) vtalvgpngsgkstvaallqnlyqptggqllld-------------gkplpqyehrylhrqv (93   )
MSBAMO (  371) tvalvgrsgsgkst-------------iaslitrfydidegeilmdghdlreytlaslrnqv (419  )
  LMRA (  371) iiafagpsgggkstifsllerfyqptageitig-------------gqpidsvslenwrsqi (419  )

A_1B0U (   92) tmvfqhfnl----------------------------------------------------- ( gap )
A_1F2U (   87) rflkgyssgeihamkrlvgnewkhvtepsskaisafmeklipyniflnaiyirqgqidaile (148  )
A_1F3O (   82) gfvfqqfnl----------------------------------------------------- ( gap )
A_1G29 (   84) amvfqsyal----------------------------------------------------- ( gap )
   TAP (   94) aavgqepqvfgrslqeniaygltqkptmeeitaaavksgahsfisglpqgydtevdeag--- ( gap )
MSBAMO (  420) alvsqnvhlfndtvanniayarteqysreqieeaarmayamdfinkmdngldtvigeng--- ( gap )
  LMRA (  420) gfvsqdsaimagtirenltyglegnftdedlwqvldlafarsfvenmpdqlntevgerg--- ( gap )

A_1B0U ( gap  ) -------wshmtvlenvmeapiqvlglskhdareralkylakvgideraqgkypvhlsggqq (155  )
A_1F2U (  149) s|alareaalskigelaseifaeftegkysevvvraeenkvrlfvvwegkerpltflsgger (209  )
A_1F3O ( gap  ) -----ipllltalenvelplifkyrga|sgeerrkraleclk|aeleerfanhkpnqlsggqq (145  )
A_1G29 ( gap  ) ---------yphmtvydniafplklrkvprqeidqrvrevaellgltellnrkprelsggqr (145  )
   TAP ( gap  ) ------------------------------------------------------sqlsggqr (160  )
MSBAMO ( gap  ) ------------------------------------------------------vllsggqr (486  )
  LMRA ( gap  ) ------------------------------------------------------vkisggqr (486  )

A_1B0U ( gap  ) ------qrvsiaralamepdvllfdeptsaldpel-vgevlrimqqlaeegktmvvvthemg (210  )
A_1F2U (  210) ialglafrlamslylageislllildeptpyldeer-rrklitimerylkkipqvilvshdee (270  )
A_1F3O ( gap  ) ------qrvaiaralannppiiladeptgaldsktgeki|qllkklneedgktvvvvthdin (200  )
A_1G29 ( gap  ) ------qrvalgraivrkpqvflmdeplsnldaklrvrmraelkklqrqlgvttiyvthdqv (201  )
   TAP ( gap  ) ------qavalaralirkpcvlilddatsaldansqlqveqllyesperysrsvllitqhls (216  )
MSBAMO ( gap  ) ------qriaiarallrdspilildeatsaldtes--eraiqaaldelqknrtslviahrls (540  )
  LMRA ( gap  ) ------qrlaiaraflrnpkilmldeatasldses--esmvqraldslmkgrttlviahrls (540  )
```

Alignment optimisation

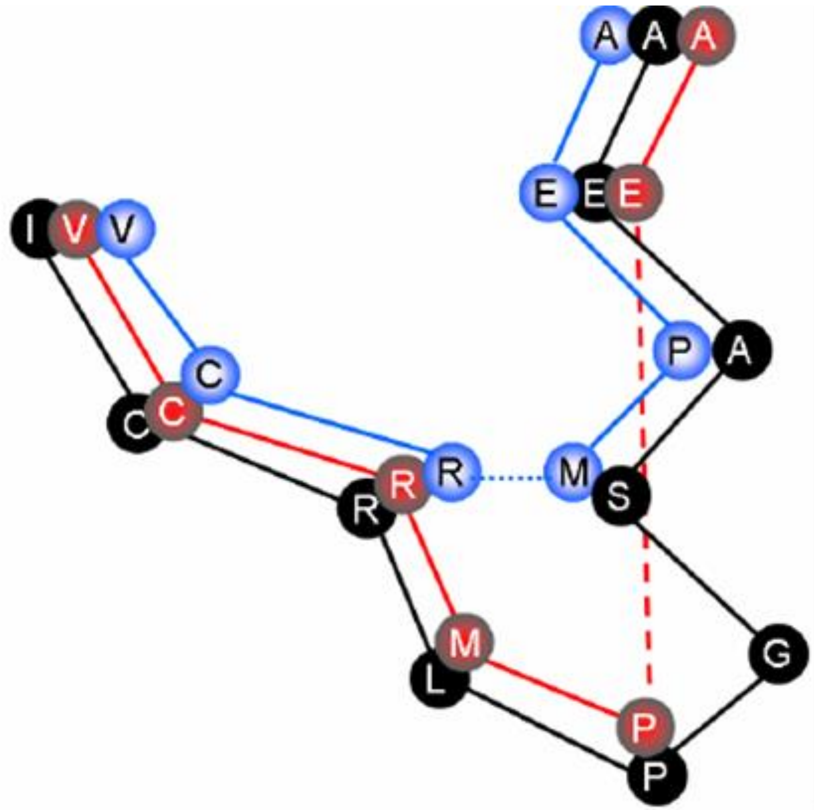
Example: Which is not the right alignment?

Sequence 1: ICRLPGSAEA
Sequence 2:VCRMPEA

I C R L P G S A E A	**or**	**I C R L P G S A E A**
VC R MP - - - -E A		**VC R - - - M P E A**

Alignment optimisation

Solution: Have a look at the 3D structure of the template!



Black: template
Red: model 1
Blue: model 2

Alignment

- **By aid of algorithms**

- **Rating of alignments by scores**
  - Sequence identity 0 – 1
  - Score-Matrices
    - PAM (point accepted mutation per 100 residues)

    → probability of an AA to mutate – PAM1: after 1 cycle, PAM250: after 250 cycles
    - BLOSUM (block substitution matrix)

Backbone generation

For aligned amino acids: backbone coordinates of the target amino acids are copied from the template.

Ramachandran Plot

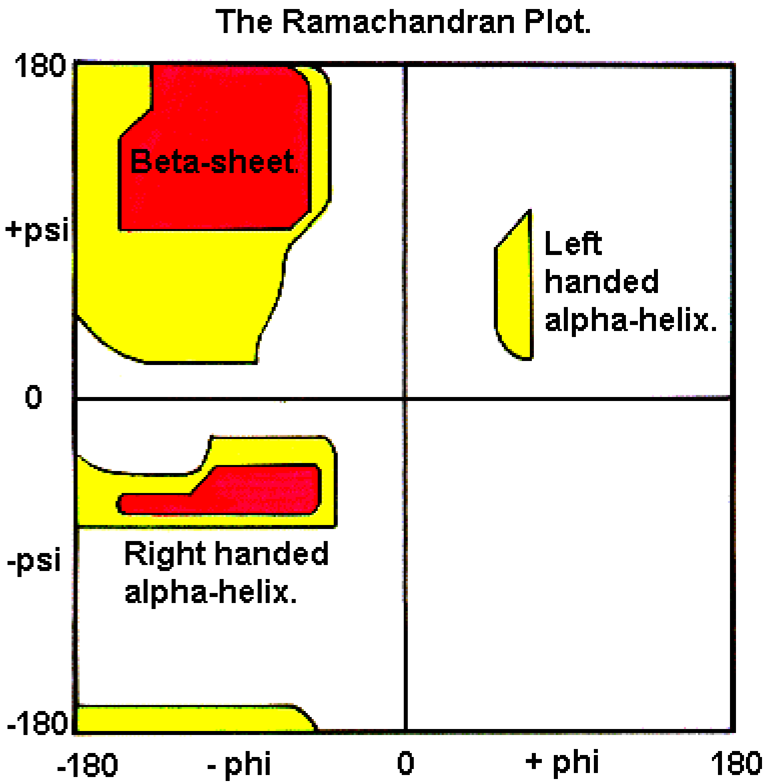
Φ (phi): angle around the α-carbon—amide nitrogen bond

ψ (psi): angle around the α-carbon—carbonyl carbon bond

Treat the atoms as hard spheres

The distance between two atoms should not be less then sum of their van der Waals radii.

phi and psi angles which cause spheres to collide correspond to sterically disallowed conformations of the polypeptide backbone

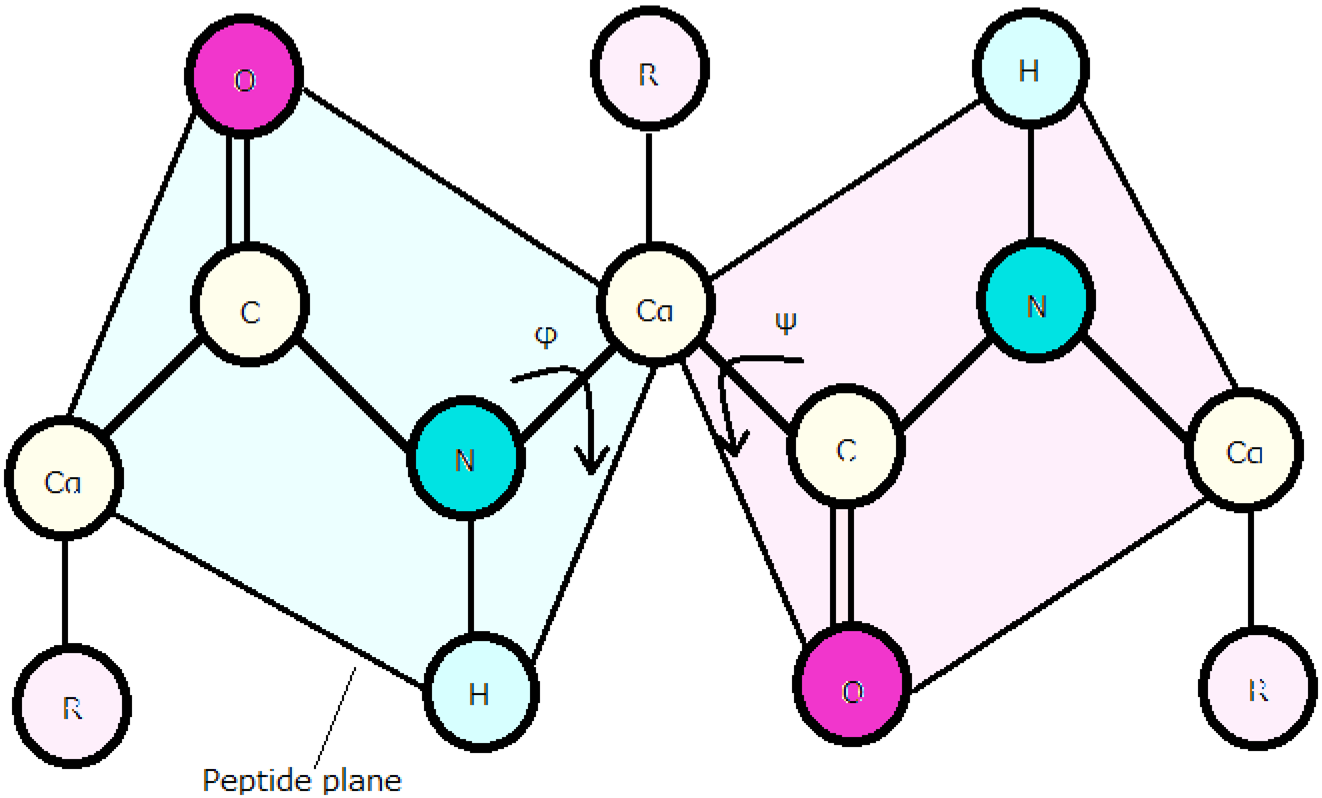


white ; disallowed regions: conformations where atoms in the polypeptide come closer than the sum of their van der Waals radii

Red; Allowed regions conformations where there are no steric clashes

yellow; show the allowed regions if slightly shorter van der Waals radi are used in the calculation, i.e the atoms are allowed to come a little closer together.

Dihedral Angels

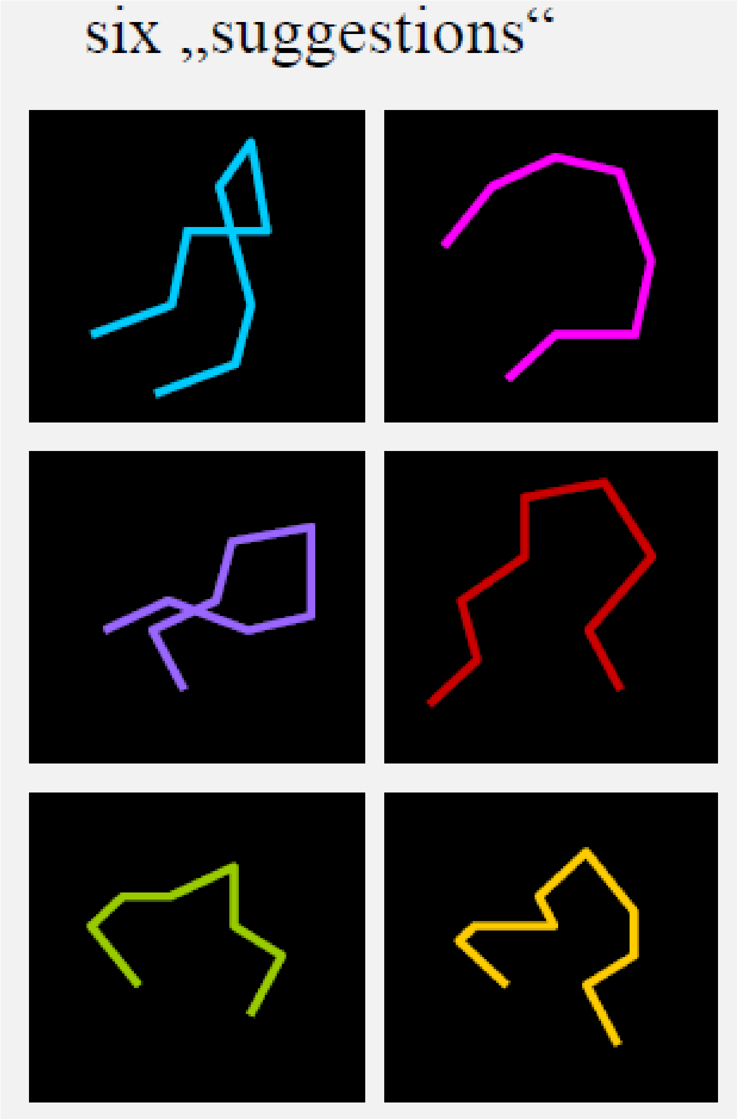


Loop modeling

Gaps in the alignment: change of structure in the backbone

- Change of structure is normally seen in loops and turns, not in regular secondary structure elements (helices and beta sheets)
- Mutations in loops (small to large side chain: glycine ; proline) can lead to different loops in template and target
- Different loop modeling methods available

LOOP MODELLING

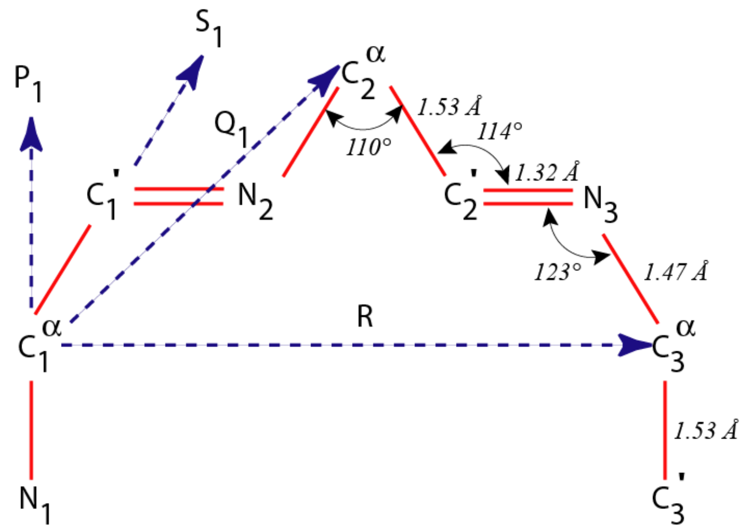


Loop modeling

- **Ab initio methods have recently received increased attention in the prediction of protein loop**
- **Potential energy function**

  Molecular mechanics force field is usually better than statistical potential in protein loop modeling.

- **Recent progress**
  - Dihedral angle sampling
  - Clustering
  - Select representative structures from ensembles



Ab initio

Ab initio: “From the beginning”

Simulates the folding of a protein

Assumption

All the information about the structure of a protein is contained in its sequence of amino acids

The structure that a (globular) protein folds into is the structure with the lowest free energy

The native structure is contained in the search space

Finding native-like conformations require

A scoring function (potential)

A search strategy

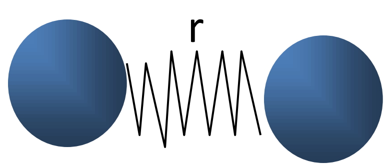
Energy Minimization (Theory)

Treat Protein molecule as a set of balls (with mass) connected by rigid rods and springs
Rods and springs have empirically determined force constants
Allows one to treat atomic-scale motions in proteins as classical physics problems

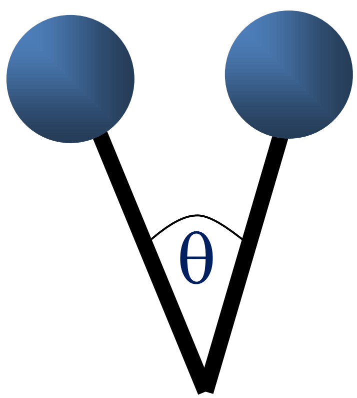
$$E_{FF} = E_{str} + E_{bend} + E_{oop} + E_{tors} + E_{vdW} + E_{el}$$

Standard Energy Function

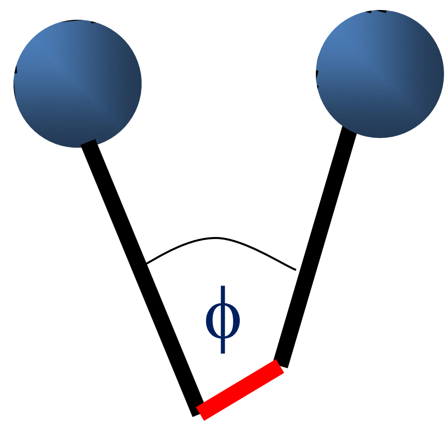
$E = K_r(r_i - r_j)^2 +$	Bond length
$K_q(q_i - q_j)^2 +$	Bond bending
$K_\phi(1 - \cos(n\phi_j))^2 +$	Bond torsion
$q_i q_j / 4\pi r_{ij} +$	Coulomb
$\varepsilon \left[\left(\frac{r_m}{r} \right)^{12} - 2 \left(\frac{r_m}{r} \right)^{6} \right]$	van der Waals



$K_r(r_i - r_j)^2$



$K_\theta(\theta_i - \theta_j)^2$



$K_\phi(1 - \cos(n\phi_j))^2$

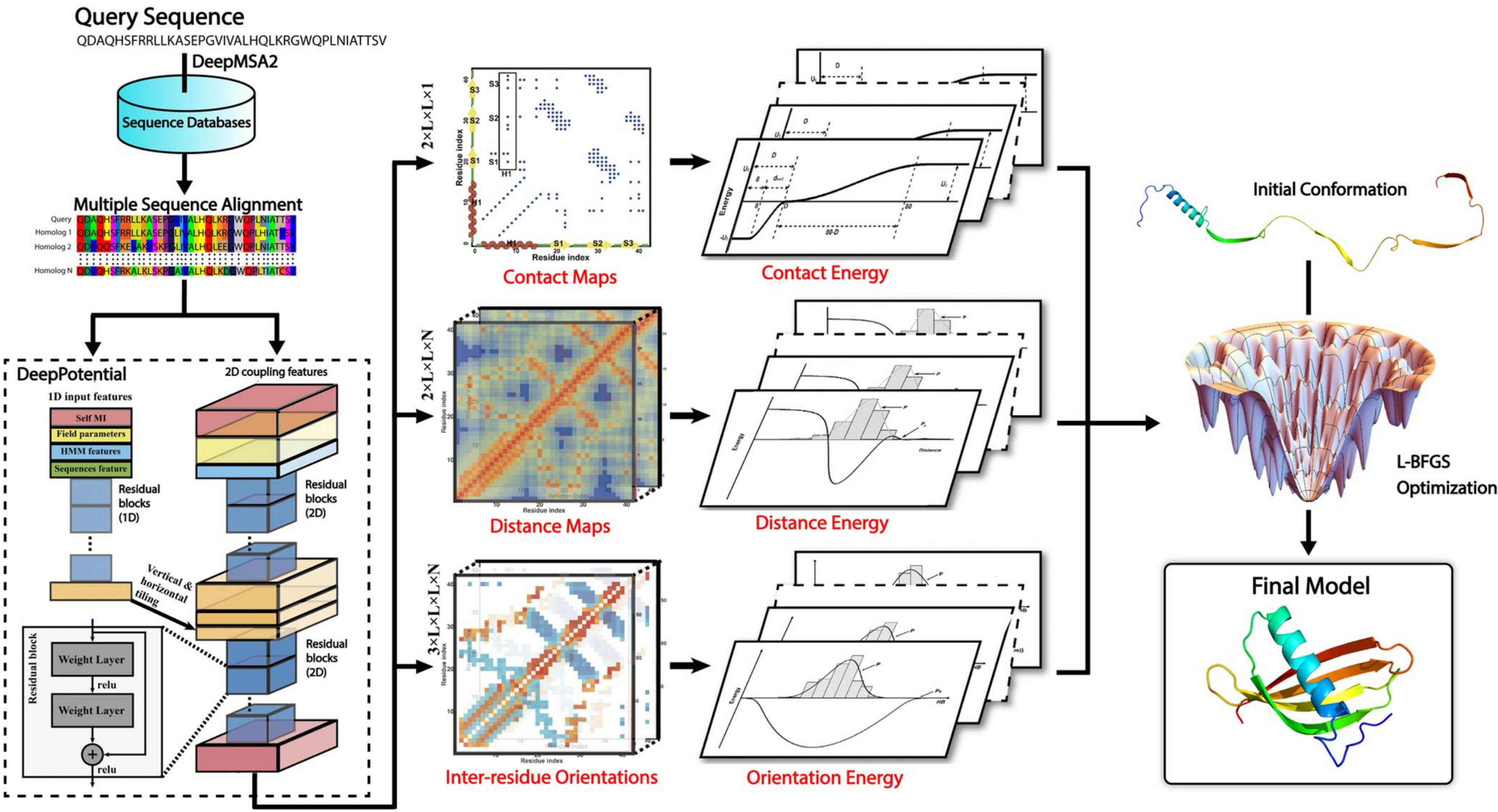
ab-initio protein structure prediction

Define some initial model

Define a function mapping structures to numerical values
(the lower the better)

Solve the computational problem of finding the global
minimum (native structure)

ab-initio protein structure prediction



Side chain modeling

- With bond lengths, bond angles and two rotable backbone bonds per residue ϕ and ψ, its very difficult to find the best conformation of a side chain
- In addition, side chains of many residues have one or more degree of freedom.
- Hence Side chain conformational search in loop regions is must
- Side chain residues replaced during coordinate transformations should also be checked
- Fortunately, statstical studies show side chain adopt only a small number of many possible conformations
- The correct rotamer of a particular residue is mainly determined by local environment
- Side chain generally adopt conformations where they are closely packed

Model optimization

- Energy minimization: removes some errors in the model (e.g. atomic clashes)
- Molecular dynamics: simulation of the motion of the protein on a femtosecond (10-15 s) timescale (Next Course)

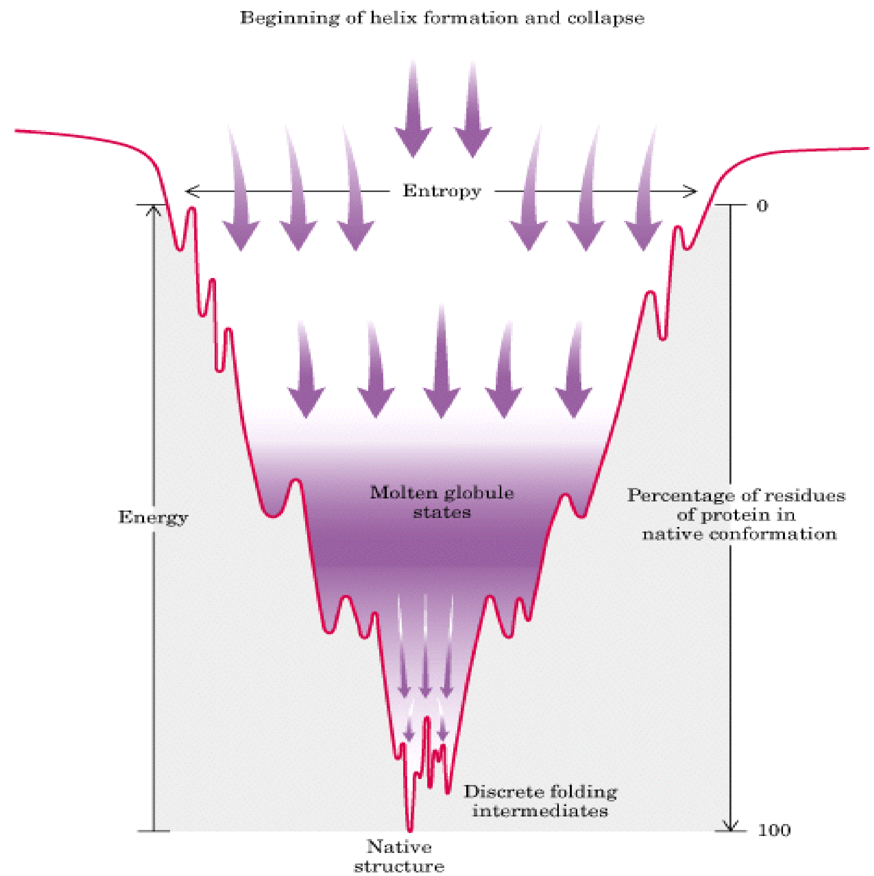
Folding funnel

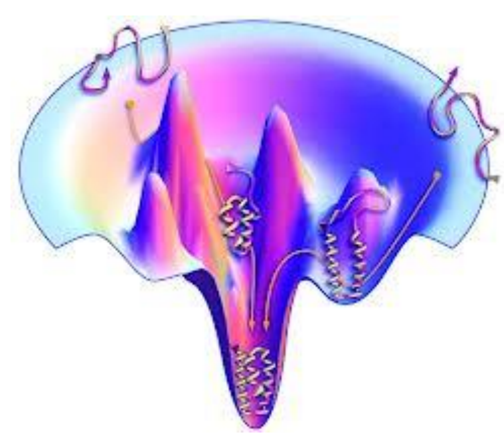
A new view of protein folding suggested that there is no single route, but a large ensemble of structures follow a many dimensional funnel to its native structure

Progress from the top to the bottom of the funnel is accompanied by an increase in the native-like structure as folding proceeds

Unfolded structures lie around the top. As the protein folds, it falls down the wall of the energy funnel to more stable conformations

The native, folded structure is at the bottom

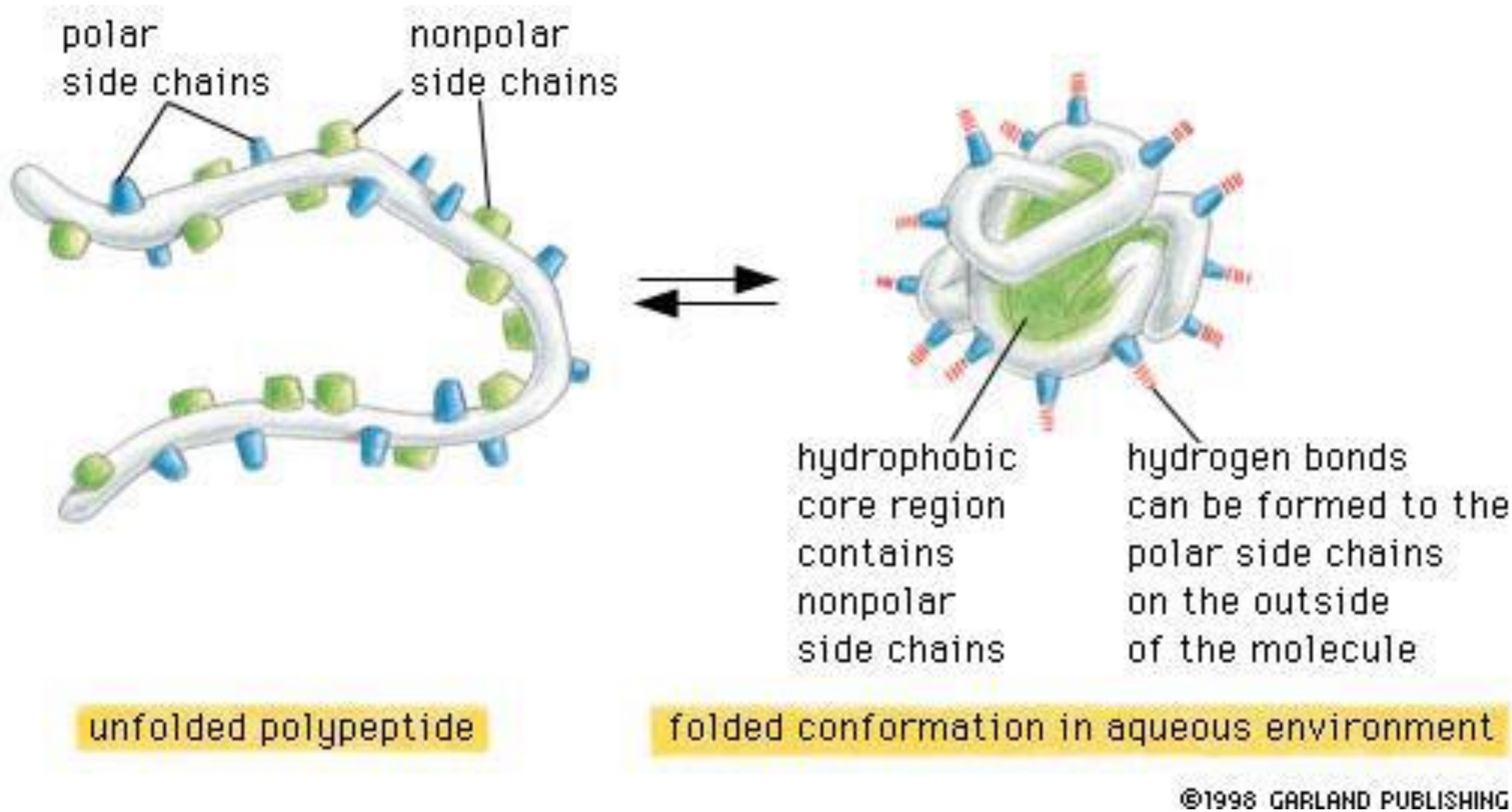




Nature Structural Biol. **4**, 10-19 (1997).

Protein Folding

Objective: **To determine the structure of a new sequence by finding its best “fit” to some fold in a library of structures**



Folding (Threading)

• Can be used if there is no protein structure with high sequence identity available

• The sequence is fitted into a known 3D fold

• SCOP -**Structural Classification Of Proteins database clustering PDB entries according to their structure (http://scop.mrc-lmb.cam.ac.uk/)**

• Even proteins having a low sequence similarity can show similar folds.

• **Scoring of the quality of the fit**

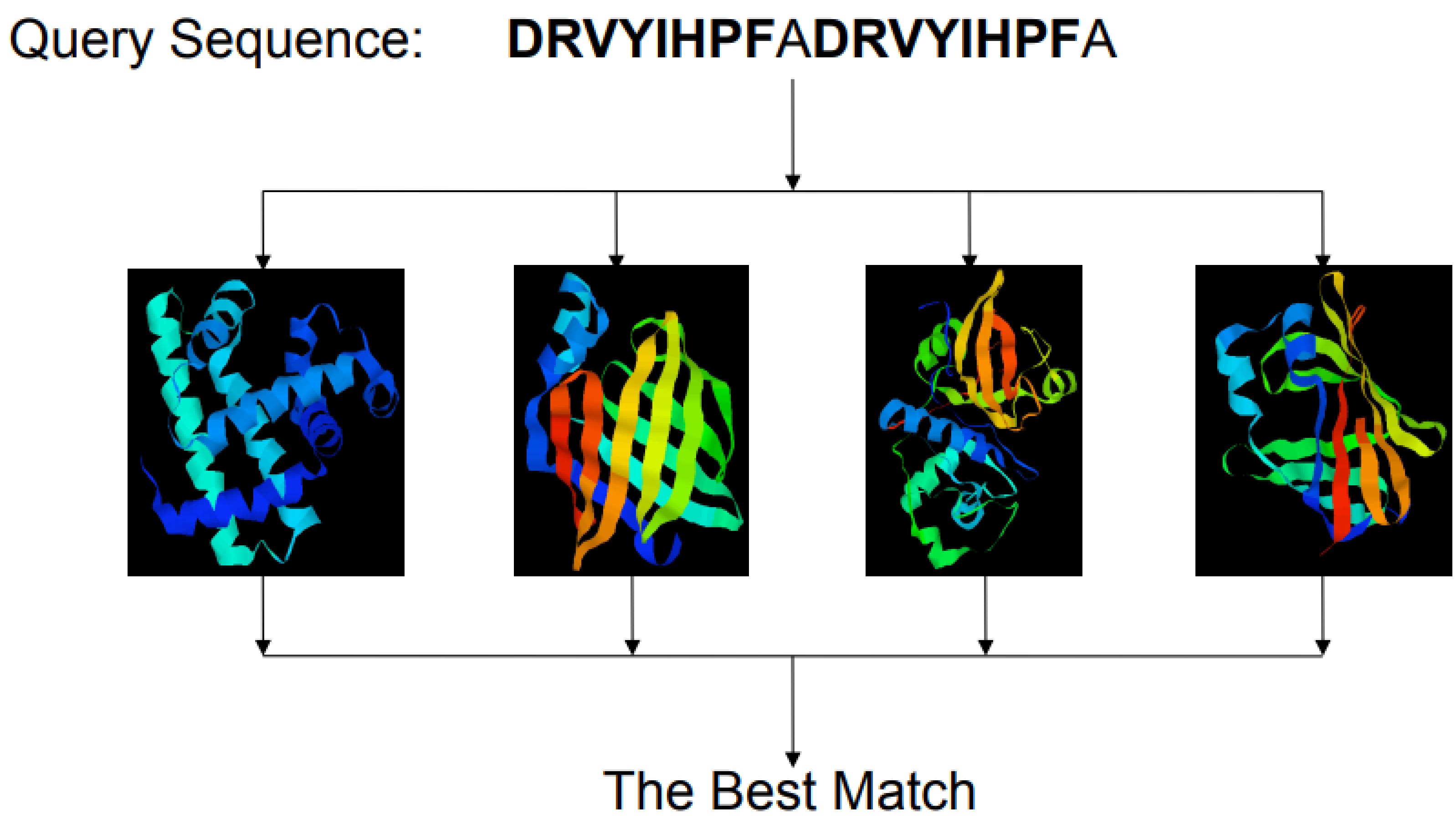
The idea was to physically "thread" a sequence of amino acid side chains onto a backbone structure (a fold) and to evaluate this proposed 3-D structure using a set of pair potentials

http://scop.berkeley.edu/

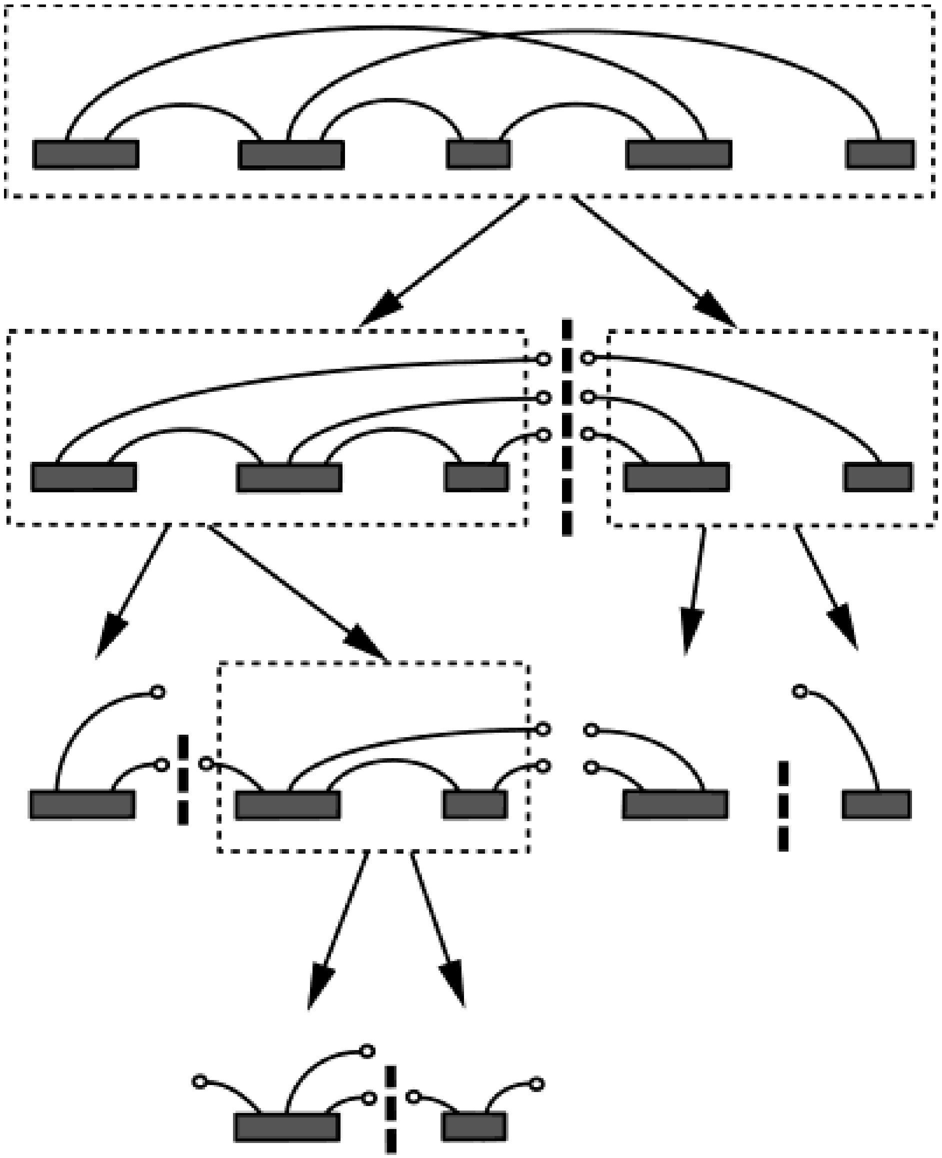
Main Steps

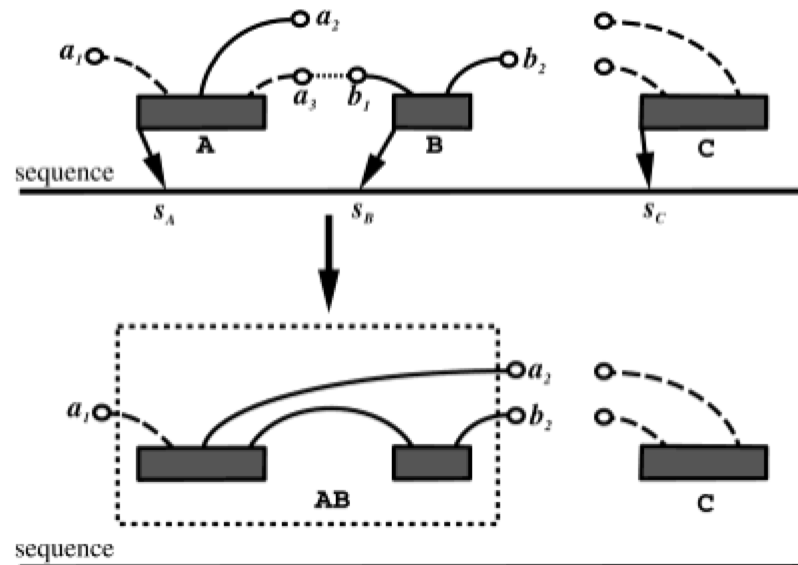
1. **Construction of Template Library**
2. **Design of Scoring Function**
3. **Alignment**
4. **Template Selection and Model Construction**

Protein Threading

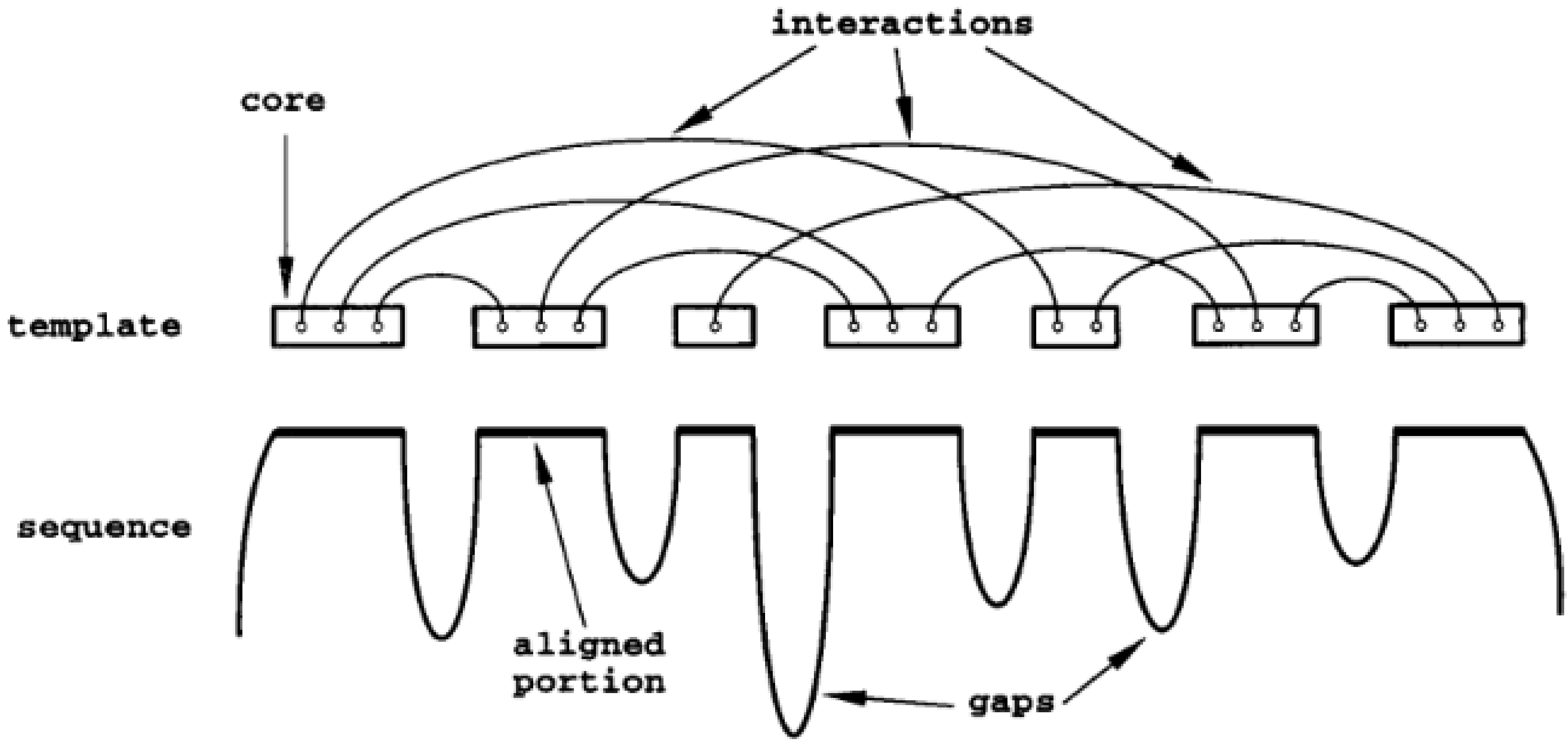


Protein Threading: The Concept

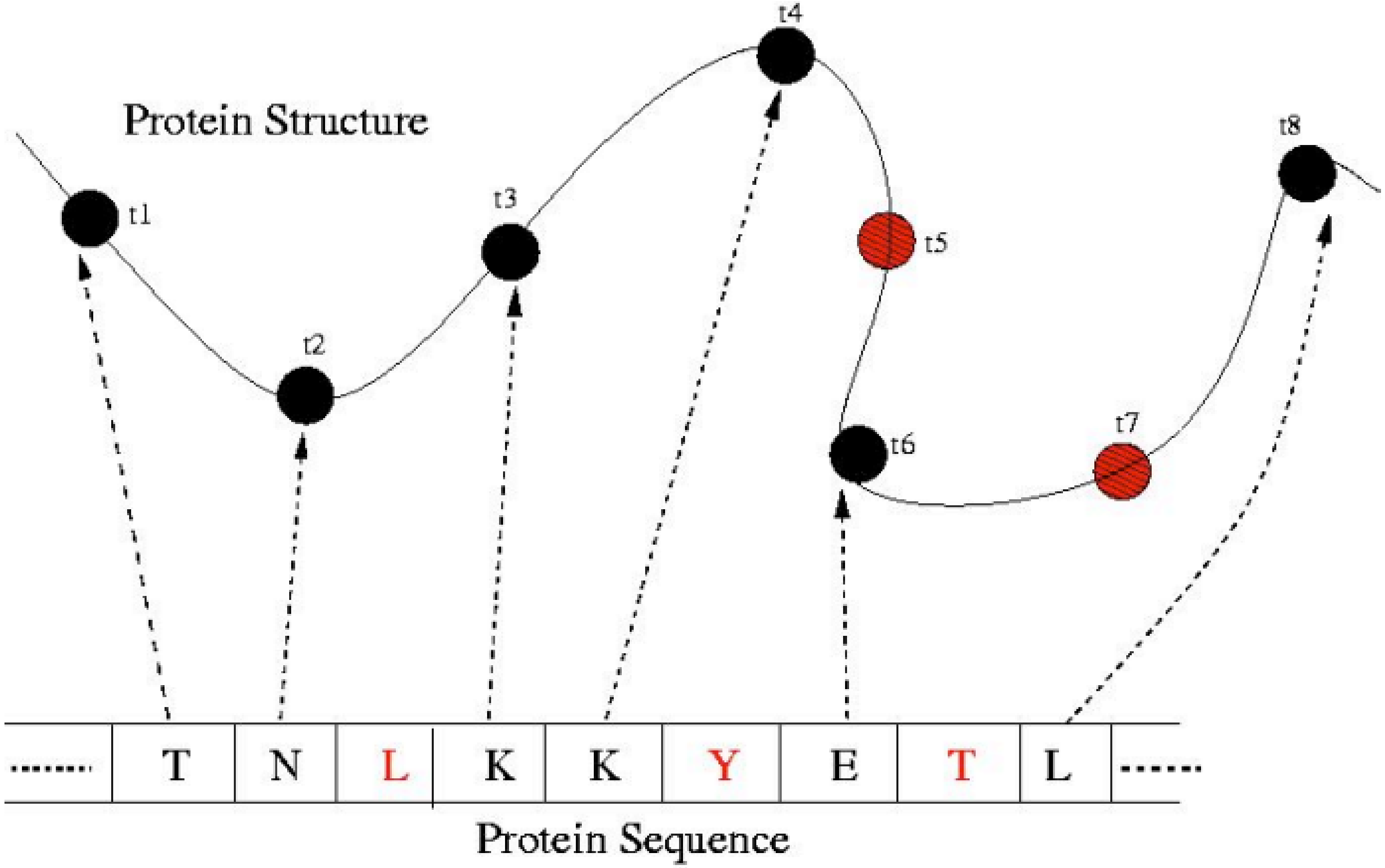




Protein Threading: The Concept



Protein Threading: The Concept

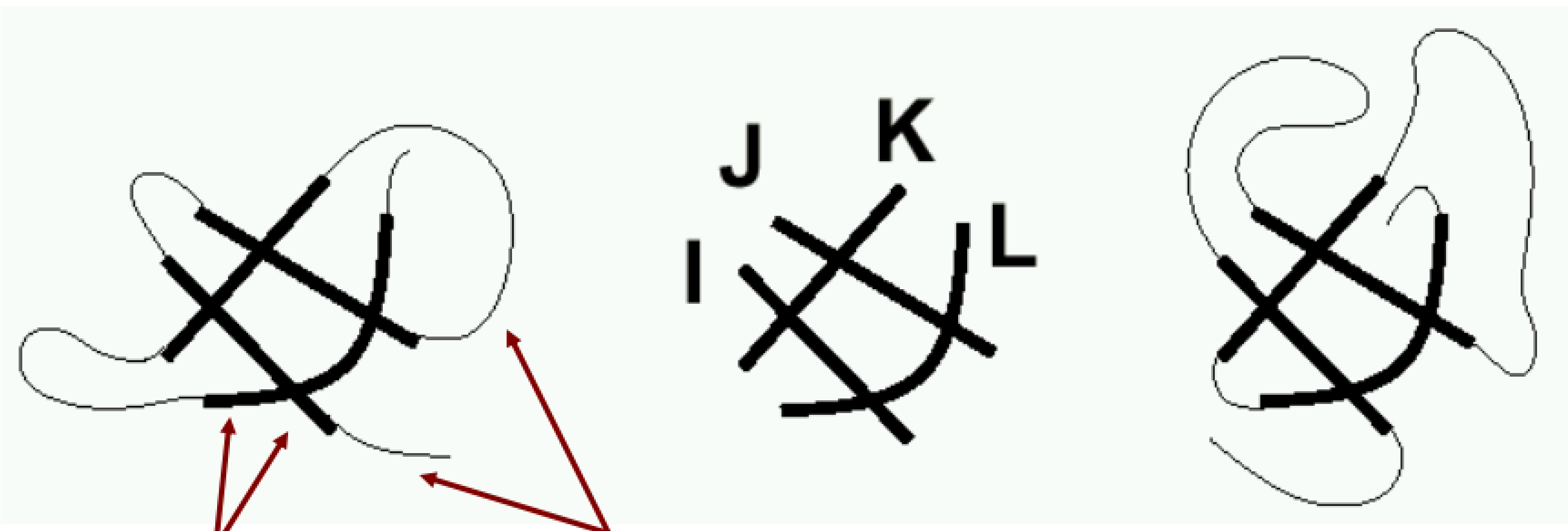


Positions or residues in red are gaps

Protein Threading: The Assumptions

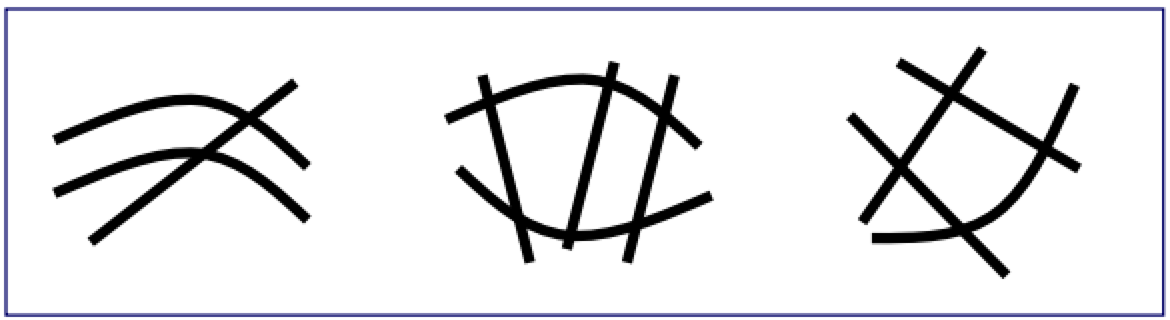
- Each template is described as a chain of cores.
- Two adjacent cores are connected by a loop.
- Cores are the most conserved segments in a protein.
- No gap allowed within a core.
- Only the pairwise contact between two core residues are considered because contacts involved with loop residues are not conserved well.
- Global alignment employed

Folding (Threading)

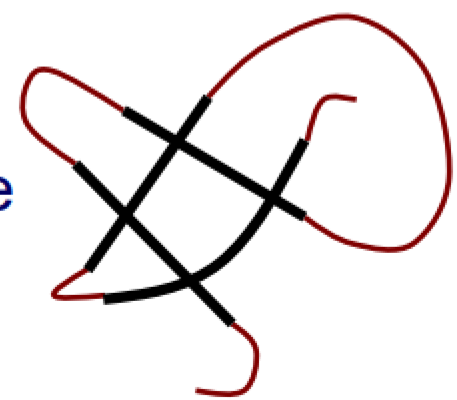


Folding (Threading)

- Given:
  - a protein sequence
  - a library of core templates

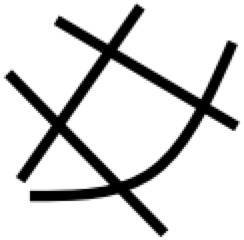


- Return: the best alignment of the sequence to a template

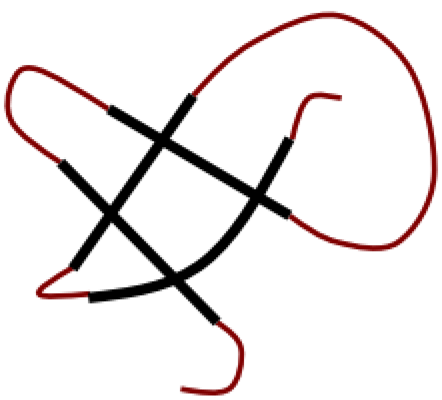


Folding (Threading)

- Given:
  - a protein sequence
  - a single template



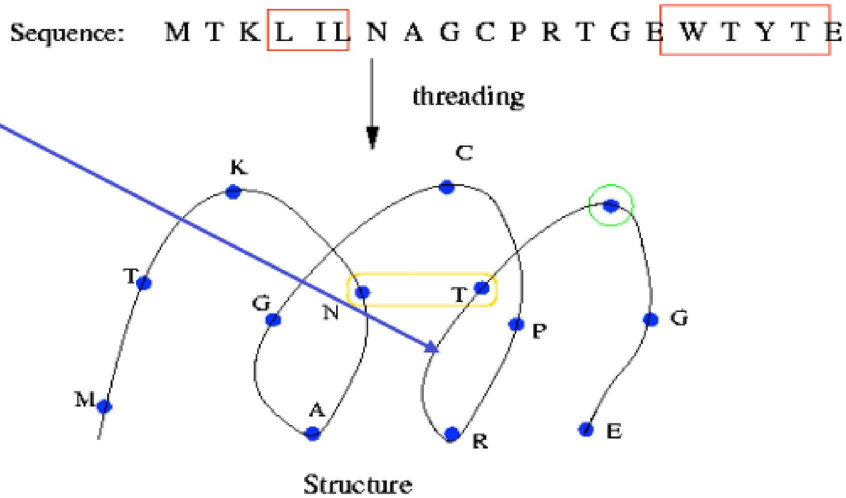
Return: the best alignment of the sequence to the template



Scoring Function

how preferable to put two particular residues nearby: E_p

(**Pairwise potential**)



how well a residue fits a structural environment: E_s

(**Fitness score**)

alignment gap penalty: E_g

(**gap score**)

sequence similarity between query and template proteins: E_m

(**Mutation score**)

How consistent of the secondary structures: E_ss

E= E_p +E_s +E_m +E_g +E_ss

Minimize E to find a sequence-template alignment

Fitness Function

occurring probability of amino acid a with s

$$FitnessScore(a,s) = -\log \frac{P(a,s)}{P(a)P(s)}$$

occurring probability of amino acid a

occurring probability of solvent accessibility s

It takes into account factors such as – amino acid preferences for solvent accessibility – amino acid preferences for particular secondary structures – interactions among spatially neighboring amino acids

Pairwise Potential

occurring probability of a and b with distance < cutoff

$$PairwisePotential(a,b) = -\log \frac{P(a,b)}{P(a)P(b)}$$

occurring probability of amino acid a

occurring probability of amino acid b